Agentic CV–JD Ranking for Automation Test Roles: Final Report

NLP501 · Group 2 · Sơn (problem and criteria) · Vũ (literature) · Đán (agentic HR, gaps, and the `coding` branch) · Hoa (datasets and metrics) · Thịnh (architecture and experiments)

Every number in this report is generated by a command in §11 and can be checked in the `docs/*_RESULTS*.md` file named next to it. The official numbers come from **NLP501-bench v1**, the benchmark the team agreed on and froze (`docs/BENCHMARK.md`).

Abstract

We built an agent that ranks Automation Test Engineer CVs against a job description and explains every rank. We evaluated it against the three questions that matter most for an agentic NLP system:

1. Is it genuinely agentic?
2. Is the data processed well?
3. How is what the LLM produces controlled?

The team froze one benchmark that mixes our 50-CV dataset with Đán's independently built 200-CV dataset: 18 JDs, a locked test split of 10, and every input pinned by checksum. On that test split:

- **The merged system is the best version of the project:** 0.847 nDCG@10 for the full agent and 0.848 for its fixed pipeline. That is ahead of Đán's original core (0.810), our v2 (0.751, p = 0.004 for the deterministic merge) and every retrieval baseline (0.68–0.73). Its lead over Đán's core is not yet significant (p ≈ 0.2).
- **Rules score; the LLM controls and explains.** The agent and its fixed twin rank alike (0.847 vs 0.848, p = 0.47). The agent's value is the work rules cannot do: asking HR on a vague JD, reading CVs the parser cannot, catching contradicted claims (2.1% → 0.6%) and keeping the arbiter to true near-ties.
- **Evaluation found and fixed real faults:**
- a recall funnel sized for 50 CVs that failed at 200 (the agent rose from 0.60 to 0.77–0.78 on those JDs);
- an arbiter acting beyond its mandate;
- a 100% name leak in masking;
- six more defects in data processing.
- **A stronger model needed more control, not less.** With Claude reading JDs, the LLM-only parser lost 0.039 nDCG@10.
- **Every LLM output is checked by code** under 13 written criteria. **The ranking ignores protected attributes:** 0 of 1,800 counterfactual pairs changed.

1. Introduction

HR teams hiring automation testers must judge three things that keyword ATS cannot see:

- **Depth:** does the candidate run scripts (L1), write them (L2), build a framework (L3) or design test architecture (L4)?
- **Use vs mention:** was a tool used, or only listed?
- **Hedging:** is a mention hedged ("familiar with", "currently learning")?

Keyword matching rewards stuffing. An LLM judge is opaque, position-biased and can invent evidence. We rank with

`Score(CV, JD) = 1[H(c)] × Σ_k w(c, k) · S(c, k)`

where:

- H is a hard knock-out on must-have tools;
- w are per-company-type weights over 8 criteria;
- S are 0–100 criterion scores.

Contributions:

1. A LangGraph agent with 12 nodes and 6 bounded loops:
2. rules set every score;
3. the LLM explains and decides only what rules cannot;
4. humans own the decision.
5. A deterministic, bilingual CV/JD NLP pipeline, measured against external gold, real text and counterfactual twins.
6. 13 written criteria for AI output (Q1–Q13), each with a code check, a target and an action on failure.
7. **NLP501-bench v1**, the team's frozen benchmark, with a head-to-head of the three versions of the project.
8. The merge of two independently built systems (ours and Đán's `coding` branch) into one.

Research question	Answered in
RQ1 Does structured extraction beat raw-text matching?	§5.1
RQ2 Do HR-configured must-haves and weights fit the company better?	§5.4
RQ3 How much does hybrid retrieval + reranking add over keyword or dense alone?	§5.1
RQ4 What does agentic orchestration add over a fixed pipeline with the same tools?	§3, §5.2, §6
RQ5 Do results and errors differ by JD type and ambiguity?	§5.1, §8

2. Data, processing and the benchmark

2.1 Data

	ours (`data/`)	coding (Đán's branch, commit `6991c3d`)	real text
Size	50 CVs × 12 JDs	200 CVs × 6 JDs	217 real, anonymised CVs
Built by	hand-written synthetic, designed to stress the pipeline (Vietnamese, long, hedges, title inflation)	template generator with known latent facts; 37% Vietnamese; 7 OCR-garbled CVs; fake PII	datasetmaster/resumes (MIT), Faker half excluded
Labels (0–3)	silver: two Claude annotators, κ_w 0.919	rubric gold (`gold_spec`)	none (NLP checks only)

No CV belongs to a real candidate without their consent. We checked every candidate public dataset. None combines real, licensed, *labelled* test-automation CVs (`docs/REALTEXT_RESULTS.md`). Consented CVs from real testers are being collected under `docs/labeling/CONSENT.md` .

2.2 NLP501-bench v1 (`docs/BENCHMARK.md`)

The two datasets were built in parallel, and each defined "relevant" its own way, so each system looked best at home. The team therefore agreed on one benchmark with fixed roles:

	dev (tuning allowed)	test (locked; the official number)
ours	JD-001, 002, 003, 006, 008, 012	JD-004, 005, 007, 009, 010, 011
coding	ambiguous_junior, mobile_offshore	ambiguous_senior, bank_api_qa, fpt_automation_test, startup_framework

How it works: - Each JD ranks the CVs labelled for it. - Our JDs were split by company type with seed 501; Đán's keep the split he pre-registered. - A manifest pins 268 input files by SHA-256 plus the coding commit, and a test fails if any of them changes. - Changes to the system are decided on dev and reported on test.

Disclosure: results on these JDs were seen during development, before the split existed. Two fixes in §5.3 were motivated by test-split errors, decided on dev, and are re-reported on test.

2.3 Pipeline: raw CV → facts (deterministic, 0 tokens)

The pipeline runs these steps in order:

1. **Normalise:** NFC, bullets, wrapped lines.
2. **Mask** contact details and protected attributes.
3. **Detect** sections and dated roles.
4. **Chunk structurally:** one chunk per role or project, labelled with its section.
5. **Find skill mentions** with the gazetteer, with NegEx-style hedge scope and evidence strength (used / listed / hedged).
6. **Compute years** from merged date ranges.
7. **Estimate automation depth** from the candidate's own action verbs.
8. **Output the `CVProfile` .**

JDs are parsed into must-haves with *one-of* groups, nice-to-haves, years, depth, languages and domains. Every fact keeps the CV line it came from (`docs/NLP_PIPELINE.md`).

Masking is deletion, not instruction. Asking an LLM to "ignore gender" is a hope; deleting it before indexing is a guarantee.

2.4 How well the processing works

Check (source)	Result
Tool extraction vs the coding generator's truth, 200 CVs (`EXTRACTION_RESULTS.md`)	F1 0.978; clean CVs 0.994; OCR-garbled CVs 0.192
Years of experience · automation depth	MAE 0.79 years · 89% exact
Redaction leak, every PII field	0%
Real CVs (217): parsed · prose precision (`REALTEXT_RESULTS.md`)	217/217 · 90%
Counterfactual twins: name / gender / age / marital status (`ROBUSTNESS_RESULTS.md`)	0 of 1,800 (CV, JD) pairs changed score

What measuring found and we fixed. Unit tests had passed on all of these:

1. **The candidate's name leaked in 100% of CVs whose first line is the bare name.** It now leaks in 0%.
2. **A gender value under an OCR-garbled label leaked.**
3. **The garbled-text flag caught 0 of 7 garbled CVs.** It now catches all 7.
4. **Double-spaced CVs lost their role titles,** which inflated one candidate's automation years from 8.0 to 14.8.
5. **"pull requests" counted as the *requests* library,** giving false credit to our own trap CV.
6. **Jira counted as automation work.**
7. **Tool names missing from the gazetteer,** found in the coding set and in real CVs.

Chunking (RQ1). With the cross-encoder, structural chunks are slightly best on both sets (0.829 / 0.628 vs whole CV 0.818 / 0.620, not significant). Without it, whole CVs retrieve best. The large gain is elsewhere: **rule scoring over the parsed profile adds +0.052 / +0.164 nDCG@10 over the same retrieval** (both $p < 0.001$; `EVAL_RESULTS*.md`, all JDs).

3. Architecture: where the agent decides

The system is a LangGraph state machine with 12 nodes and 6 bounded loops (`docs/architecture.html`). Its rule: **rules where facts exist, the LLM where judgement or reading is needed, humans where the decision is theirs.**

Loop / node	Decision	Decided by
L1 clarify	pause and ask HR, or proceed	rules first (no must-have tools; title contradicts years), then a typed LLM gate
L2 expand	widen the search; which related tools may substitute	coverage rule proposes, **HR approves** each tool
L6 investigate	which tool to call next on an unreadable CV, and what it shows	**LLM** (typed actions); **code** verifies every claim
L4 verify	retry a review whose quotes or claims fail the checks	code (Q2–Q5)
L3 arbiter	swap a *near-tie* (	Δ
L5 HR review	approve, re-weight, or approve/reject candidates	**HR**; weights re-learned from decisions (pairwise logistic regression, audit log)

Why this is agentic rather than a linear pipeline. The path depends on what the agent observes. Across the 18 benchmark JDs the agent took 8 distinct paths; its fixed twin took 1. The investigator (N12) is a tool-using ReAct loop in the strict sense: the model chooses its next action from what the previous one returned. Code then decides whether the result becomes a fact.

Why the LLM does not score. In v2 the LLM scored 5 of 8 criteria, and the agent then ranked 0.033 below its own deterministic input. Since v3:

- rules set all 8 scores;
- the LLM's judgement is a verified review;
- the agentic parts do what rules cannot: ask, read noisy text, break ties, explain.

4. Controlling what the LLM produces

`docs/AI_OUTPUT_CRITERIA.md` defines 13 criteria. Each has a code check, a target and an action on failure.

	Criterion	Check	Action on failure
Q1	schema-valid output	pydantic	retry, then safe default / needs review
Q2	quotes grounded in the CV	fuzzy span match	L4 retry naming the failing quotes
Q3	quotes support their criterion	right section and entity	labelled "listed, not demonstrated"
Q4	rationales consistent with the parsed CV	no tool or years the CV does not show	L4 retry with the contradiction, then `[unverified]`
Q5	no protected attributes	EN/VI lexicon	sentence removed
Q6	report claims supported per candidate	Q4 + Q5 per candidate, no foreign ids	regenerate once, then drop failing sentences
Q7	arbiter not position-biased, near-ties only	both orders;	Δ
Q8	decisions right	planted scenarios	safe default
Q9	explanation adds up to the score	by construction	—
Q10	ranking agrees with people	benchmark metrics with CIs	not adopted
Q11	ranking ignores protected attributes	counterfactual twins	release blocker
Q12	bounded cost	calls, tokens, latency	loop limits
Q13	an LLM judge is validated before use	κ vs humans	not used as ground truth

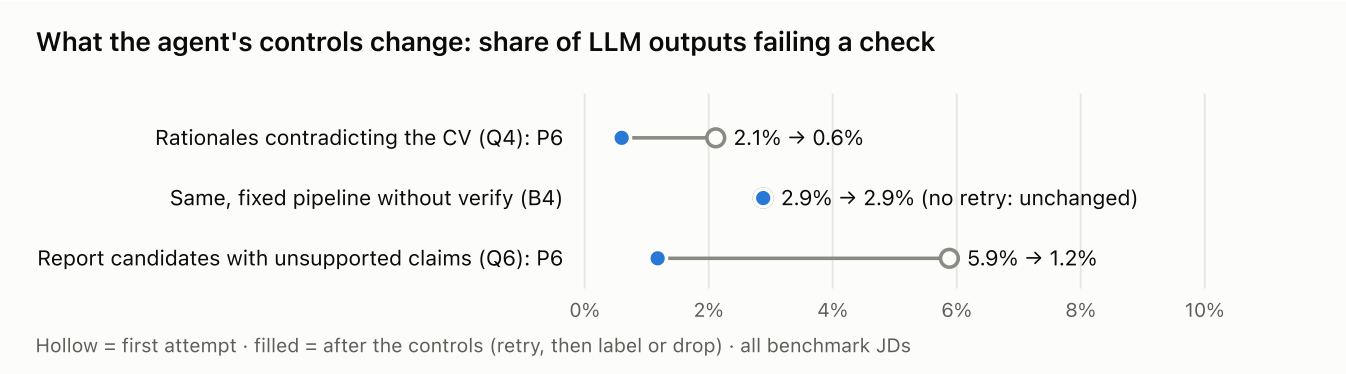


Figure 1. The share of LLM outputs failing a check, on the first attempt (hollow) and after the graph's controls acted (filled), across all benchmark JDs. Numbers: the AI-output scorecards in `EVAL_RESULTS*.md`.

Measured on the full agent across the benchmark JDs: - **Q1:** 0 invalid outputs. - **Q2:** 100% of quotes grounded. - **Q4:** contradicted rationales 2.1% → 0.6%, against 2.9% for the fixed pipeline, which has no retry. - **Q6:** unsupported report claims 5.9% → 1.2% after one regeneration; the rest dropped, none reaching HR. - **Q7:** 100% of arbiter verdicts stable under an order swap after the |Δ| fix (§5.3); 90% before it, when out-of-mandate pairs were also judged. - **Q8 decisions** (`SCENARIO_RESULTS*.md`): - L1 gate 18/18 correct; - L6 investigator 0 false recoveries in 42 planted cases (all misspelt tools recovered, all negatives kept out); - Sonnet 5.5 recovered 9 of 16 true facts from garbled CVs against qwen3:8b's 7, both with 0 false recoveries. - **Q11 fairness:** 0 of 1,800 counterfactual pairs changed score.

Three properties make this control rather than hope:

1. **The checks are code,** so they run on every output.
2. **They are measured before and after acting.**

3. **They have regression tests.** Writing the Q2 retry path exposed a silent regression: the retry prompt never told the model what was wrong.

The checks favour precision and are a lower bound on errors; the blind human audit measures what they miss.

A stronger model needed more control. Claude Sonnet reads JDs more strictly than qwen3:8b:

- it listed 4–7 ambiguities per JD where qwen listed none;
- it gave 5 must-haves for JD-008 where qwen gave 2, turning "Jenkins or GitLab CI" into two hard requirements.

With the LLM-only JD parser, ranking fell **0.884 → 0.842** (ours) and **0.824 → 0.795** (coding), and knock-out misses rose from 9 to 17. The hybrid parser, where the deterministic parser decides and the LLM fills gaps, held at 0.881 / 0.792 under both models. It is the default, chosen by the pre-registered rule.

5. Experiments

Setup: - **LLM:** Claude Sonnet 5.5 (reasoning) and Haiku 5.5 (fast gates), served through the Claude Code CLI as a stand-in for the Anthropic API. - **Reproducibility:** JD requirements are extracted once and shared by every system, LLM responses are cached, and `PYTHONHASHSEED=0` is pinned. - **Statistics:** means over JDs with 95% bootstrap CIs; paired one-sided bootstrap p for "A > B".

5.1 The official benchmark (RQ1, RQ3, RQ5): `docs/BENCHMARK_RESULTS.md`

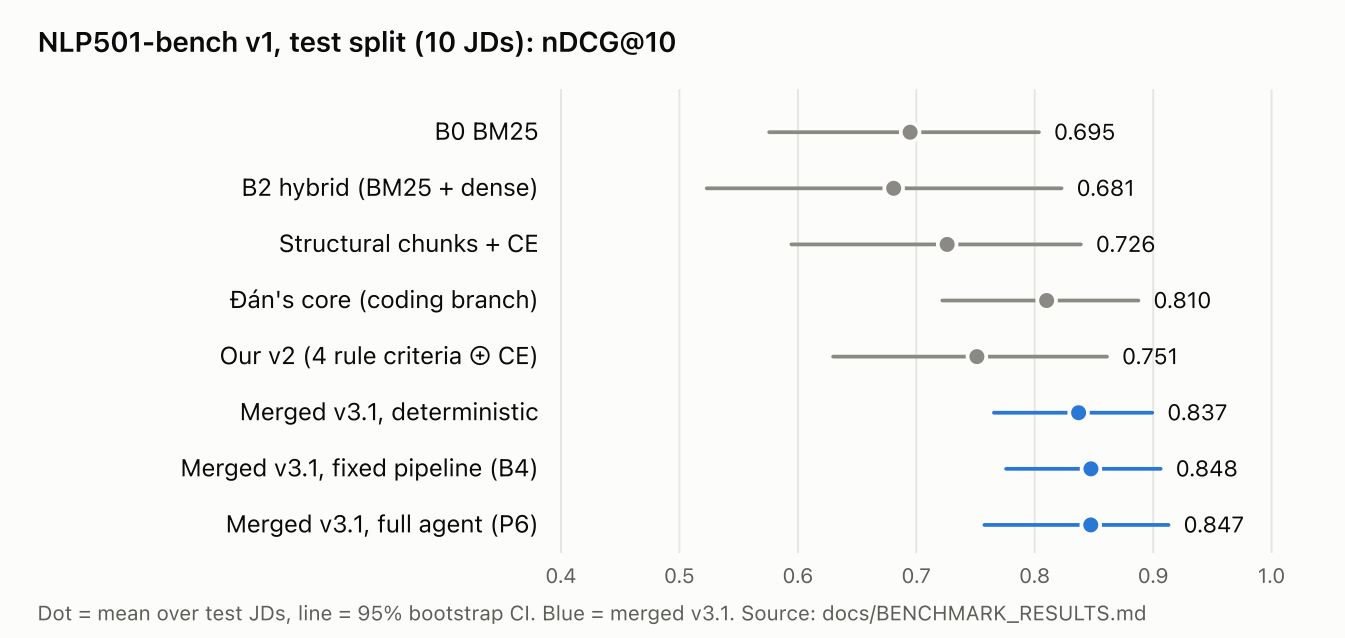


Figure 2. nDCG@10 on the 10 locked test JDs, mean and 95% CI. Blue marks the merged system.

System	test nDCG@10 (10 JDs)	ours (6)	coding (4)	P@5	dev nDCG@10 (8)
B0 BM25	0.695	0.800	0.536	0.680	—
B2 hybrid (BM25 + dense)	0.681	0.859	0.413	0.600	—
Structural chunks + cross-encoder	0.726	0.830	0.570	0.720	—
Đán's core (coding branch, unchanged)	0.810	0.773	**0.866**	0.800	0.846
Our v2 (4 rule criteria ⊕ CE)	0.751	0.806	0.669	0.700	0.805
Merged v3.1, deterministic (B3)	0.837	0.890	0.758	0.880	0.869
Merged v3.1, fixed pipeline (B4)	**0.848**	0.890	0.784	0.880	0.872
Merged v3.1, full agent (P6)	0.847	**0.897**	0.772	0.880	0.866

Which version is best. The merged system leads overall:

- **Over our v2:** +0.086, p = 0.004.
- **Over Đán's core:** +0.027 (p = 0.27) for the deterministic merge and +0.037 (p = 0.22) for the agent.

So it is the best version measured, but its lead over Đán's core is not yet significant on 10 JDs. **Each original still wins at home:** Đán's core on his JDs (0.866), and the merged system on ours (0.89). The merge combines his 8-axis rules and our NLP and retrieval, so it is the only version that is strong on both.

RQ3, retrieval. - Retrieval alone tops out at 0.73. - BM25 beats dense retrieval on template CVs, consistent with Resume2Vec's finding that testing is where keyword search wins. - The cross-encoder over requirement-wise passages is the largest retrieval gain.

RQ1, structure. Structured facts beat text by a wide margin: 0.837 against 0.726 for the best retrieval-only system on test.

RQ5, by JD. Ambiguity, not vocabulary, is where ranking breaks. The lowest test JD for every LLM-free system is `ambiguous_senior` (no concrete must-have). It also holds all of the agent's avoidable misses on Đán's set (§8).

5.2 Agent vs fixed pipeline, and ablations (RQ4)

On the 10 test JDs the full agent and its fixed twin rank alike (0.847 vs 0.848, p = 0.47). That is by design: rules set the score in both. Across all 12 of our JDs (`EVAL_RESULTS.md`):

System	nDCG@10	contradicted claims (final)	distinct paths	LLM calls
P6 full agent	0.882	0.9%	5	177
B4 fixed pipeline	0.879	4.0%	1	108
A1 – verify (L4)	0.883	2.8%	3	184
A2 – company profile	0.878	1.3%	5	197
A3 – arbiter (L3)	0.880	0.9%	5	157
A4 – investigator (L6)	0.883	0.9%	5	179

The ablations A1–A4 ran before the two fixes in §5.3. On our 50-CV pool those fixes change nDCG@10 by at most 0.003 per system (§5.3), well inside the noise.

What each part adds: - **Verify (L4)** is what removes contradicted claims. - **The arbiter** reorders true near-ties and caught position bias in its verdicts. - **The investigator** costs nothing on clean CVs and is the only part that can rescue a good candidate from noisy text (§6.2). - **The cost of control** is 1.6× the LLM calls (+54% tokens) of the fixed pipeline. Uncached, that is about 80 s per JD through the CLI against about 35 s.

5.3 What the benchmark exposed, and how it was fixed

1. A recall funnel sized for 50 CVs.

- **Symptom:** on Đán's test JDs the agent scored **0.601 (P6) and 0.608 (B4)**, far below its own deterministic core (0.758). On our JDs B4 and B3 are identical.
- **Cause:** search carried the top 40 retrieval hits into screening. That is 80% of our pool but **20%** of a 200-CV pool, and retrieval is weak on template CVs. CVs that belong in the top 10 ended up at ranks 12–99.
- **Fix:** rules cost nothing, so the top 40 CVs **by rule score** now also enter screening (`Settings.rule_recall`).
- **Decision, on dev only:** B4 ours 0.871 → 0.869 (within tolerance), coding 0.857 → 0.880.
- **Re-reported on test:** B4 on Đán's JDs **0.608 → 0.784**; `bank_api_qa` alone 0.43 → 0.84.

2. An arbiter beyond its mandate.

- **Symptom:** a real JD-003 trace showed "Near-tie (Δ = −0.150) resolved by arbiter".
- **Cause:** with fused ordering a neighbour can have the *higher* total, and the check treated every negative gap as a tie, so the LLM lifted a 0.77 candidate over a 0.92 one.
- **Fix:** the check now uses |Δ|, with a test. P6 was re-run.

5.4 Company profiles and the JD parser (RQ2)

Company profiles. Weights differ by company type. Turning the profile off (A2) moved nDCG@10 by −0.004, not significant. The silver labels were written without a company context, so they cannot show a profile effect. That limit is stated in §9; the human labels will help.

HR-taught weights. HR can re-weight in words, or by approving and rejecting candidates. The weights that best explain those decisions are learned by pairwise logistic regression and logged.

The JD parser. The parser choice mattered more than the weights: hybrid beat LLM-only by +0.039 on ours (§4).

6. Case studies

6.1 A vague JD: the agent stops to ask, and every control fires

JD-003 asks for "good automation skills" and names no tool. This is a real run (Claude), condensed from the agent's own narration (`scripts/smoke.py JD-003`):

```
[ingest]   Must-have: — · min years: None · depth: L2
[ingest]   Rule: no concrete must-have tools → ask HR (no LLM needed)
HR answers: "3+ years, Selenium or Playwright, English required"
[ingest]   Canonicalized must-have ['selenium', 'playwright'] → ['selenium'] (playwright as
accepted alternative)
[search]   → 43 candidates (3 added by rule score that retrieval missed); knock-out screened
the whole pool: 12 out (4 only hedged)
[investigate] No knock-out looks like a parsing failure → investigator not needed (0 LLM
calls)
[rerank]   kept top 8 for LLM assessment (saves 30 LLM calls)
[verify]   87/87 quotes grounded; 1/40 rationales contradict the parsed CV → CV-027 sent
back to assess
[verify]   8/8 grounded; 0/5 contradict
[score]    Data-quality flags on ['CV-047'] … every CV has a status: 8 ranked, 30 not
shortlisted, 12 knocked out
[arbitrate] Near-tie CV-033 / CV-038: the verdict flipped when the order was swapped
(position bias) → rule order kept
[report]   Report ready for top 5 candidates; every claim checked against the CVs
```

What the run shows:

1. **A rule, not the model, decided to ask HR.** There was nothing to rank fairly without a must-have.
2. **HR's "or" was canonicalised into a one-of group.**
3. **The rule channel recovered 3 CVs that retrieval missed.**
4. **The investigator correctly stayed idle.**
5. **A contradicted rationale was caught and repaired.**
6. **The arbiter detected its own position bias and deferred to the rules.**
7. **Every CV ended with a status and a reason.**

(The run above predates the near-tie fix: it also showed the out-of-mandate swap that §5.3 removed.)

6.2 A garbled CV: the investigator reads what the parser cannot

syn_091 is one of Đán's OCR-garbled CVs. The parser finds none of the fpt JD's must-haves, so the CV is knocked out. The deterministic trigger "the text is garbled" starts the investigator. Its real actions (Sonnet 5.5, `SCENARIO_RESULTS_claude-sonnet-5-5.md`):

```
read_section(summary)            → "2 y#%rs%of e@%er%@%~e in softw@re testing."
propose(playwright, quote)       → accepted: playwright (match 0.80)
search(Python)                   → No line resembles 'Python'.
search(API testing)              → "U~ed P@aywrig@t,~Webd#iverIO, Postman,#R~b~% @@amework,
T~stN…"
propose(api_testing, quote)      → accepted: api_testing
```

On another garbled CV (syn_043), the model proposed Python from a line that does not contain it. Code answered `rejected: the quote does not name python (best match 0.50)`, and the knock-out stood. **The model chooses where to look; code decides what becomes a fact.** Across 42 planted cases that rule produced 0 false recoveries.

7. Hypotheses vs results

The midterm deck stated four quantitative hypotheses before any experiment. Here they are, against what we measured.

Hypothesis (midterm)	Predicted	Measured	Verdict
H1: the agent beats the fixed pipeline on ranking	P6 86.8 vs B4 83.2 nDCG@10	test: 0.847 vs 0.848	**Not supported.** Rules score both, so ranking is equal by design. The agent's value moved to control (§4)
H2: the selective arbiter cuts token cost	−78% tokens; 7.2 s, $0.09 per JD vs 18.4 s, $0.42	P6 uses 1.6× the calls (+54% tokens) of our B4	**Refuted as stated.** The −78% compared against an LLM that reranks every candidate. Our fixed pipeline already calls the LLM only for a shortlist of 8, so the agent's loops add cost
H3: verification keeps explanations grounded	97.6% grounded (A1 72.2%)	100% grounded with and without verify	**Met, for a different reason.** Claude rarely invents quotes; verify's measurable effect is on contradicted *claims* (2.8% without it vs 0.9% with it)
H4: company profiles align with HR	τ 0.81 vs 0.52 generic	τ 0.552 vs 0.542 (A2)	**Not supported** on labels written without company context

Two refuted and two reframed hypotheses are a result. They moved the design from "the LLM decides and the agent saves tokens" to "rules decide and the agent controls the LLM".

8. Error analysis (`docs/ERROR_ANALYSIS.md`, test split)

We count only the errors that cost HR something:

- **avoidable misses:** a relevant CV ranked below 10 although every CV graded at least as high fits in the top 10;
- **false alarms:** an irrelevant CV in the top 5.

	ours (6 test JDs)	coding (4 test JDs)
P6 full agent	5 misses, 1 false alarm	5 misses, 1 false alarm
Đán's core	10 misses, 1 false alarm	4 misses, 0 false alarms

Patterns: - **On ours, 3 of the agent's 5 misses are strict knock-outs:** a good CV missing one stated must-have (Selenium once, GitLab CI twice). The other 2 are weighting, on the architect JD. - **On Đán's set, every error is on one JD, the vague `ambiguous_senior`:** all 5 misses and the false alarm. There the scripted HR answer added a web-tool must-have and 5 years, which the rubric weighs differently. Ambiguity, not vocabulary, is the remaining failure mode. - **The one false alarm on ours** is a CV whose must-have (Python) is only listed, never used.

9. Limitations and ethics

- **Silver labels.** Our labels come from two Claude models, which agree too readily (κ 0.919). Two teammates are grading a blind 120-pair sample (`docs/labeling/`). If human–silver κ ≥ 0.8 the set stays; otherwise the human labels replace it as benchmark v1.1. Until then, ranking numbers are provisional.

- **Synthetic ranking data.** Real text was used only to check the NLP. Consented real CVs are being collected. Comparisons between systems are the result; the absolute numbers will not transfer.
- **Small test split.** With 10 test JDs, CIs are ±0.05–0.08. Differences under about 0.03 are not significant.
- **The served LLM.** Claude ran through the Claude Code CLI as a stand-in for an API. Calls, tokens and behaviour carry over; latency and cost do not. The labels and the agent are both Claude, but rules set the score, so this cannot inflate ranking metrics directly.
- **Ethics.**
- Protected attributes are deleted before indexing, and the twin test proves the ranking ignores them.
- The system recommends; HR decides, and every HR decision is logged.
- No real CV is used without consent.

10. Conclusion

An agentic CV ranker earns trust by being agentic where rules cannot reach: asking, reading noisy text and judging true ties. It keeps the score in deterministic rules over well-processed data and checks every LLM output in code. Measured on one frozen benchmark, the merged system is the best version of the project. Its evaluation found faults that unit tests never would.

Next: - human labels (benchmark v1.1); - consented real CVs; - a larger test split, enough to separate the merged system from Đán's core with confidence.

11. Contributions and reproduction

Contributions: - **Sơn:** problem framing, hiring criteria and company profiles. - **Vũ:** literature review (31 papers). - **Đán:** agentic-HR analysis, gaps G1–G5, and the `coding` branch, whose JD parser, 8-axis rules, both-order arbiter, explanation cards, HR calibration and synthetic benchmark were merged. - **Hoa:** datasets, metrics and human validation of the labels. - **Thịnh:** architecture, the agent, the NLP pipeline, the benchmark and the experiments.

```
uv sync && docker compose up -d && (cd backend && uv run python -m app.rag.ingest)
cd backend && export PYTHONHASHSEED=0 LLM_PROVIDER=claude_code
uv run pytest                                    # 123 tests, no LLM or Docker needed
uv run python -m app.eval.benchmark verify       # the frozen benchmark is intact
# full benchmark run: docs/BENCHMARK.md › Reproduce;  official table: python -m
app.eval.benchmark report
uv run python -m app.eval.figures                # figures in docs/figures/
uv run python -m app.eval.extraction && uv run python -m app.eval.robustness && uv run
python -m app.eval.scenarios
uv run python -m app.eval.errors P6 "Đán core (coding branch)" --bench test
```